



Sodium-ion battery system—several of which can be rack-mounted together for large applications (Credit: <https://bridex.fujielectric.com>)

sity between 100 and 120Wh/kg, a power density between 2 and 5kW/kg, a 5-minute charging time, and a lifetime of 5000+ cycles to 80% of capacity.

Natron Energy. A California-based company, Natron, in partnership with Clarios, is going to mass-manufacture rack-mounted sodium-ion battery systems for industrial use in large data centers' backup power, EV charging stations, and other industrial applications such as forklifts. The batteries use Prussian blue analogues for both the anode and cathode with an aqueous electrolyte and can achieve an energy density of 70 watts per kilometre per kilogram. The batteries offer strong volumetric power density, an 8-minute charging time, a lifespan of over 50,000 cycles, and thermal stability for safe transportation.

Altris AB. Altris AB is a Swedish company that has developed a technology for mass production of Prussian white, known as Fennac. Fennac is considered the ideal material for use as cathodes in SIBs. SIBs utilising Fennac cathodes, along with a hard carbon anode and a patented non-flammable electrolyte, can achieve a maximum capacity of 170mAh/gm and an average voltage output of 3.2V.

The SIB bright sparks in India

India, as the largest importer of oil in the world, aims to reduce its import burden by promoting the adoption of electric vehicles that rely on LIBs. At the COP26 summit, India

committed to meeting 50% of its energy requirement from renewable sources and increasing non-fossil fuel power generation capacity to 500GW by the end of the decade. Renewable energy sources, such as wind and solar, utilise battery energy storage

systems (BESS) to ensure reliable and uninterrupted energy supply. Currently, LIBs fulfill the demand for BESS in India, with the market estimated at US\$4,300 million and expected to grow to US\$25,300 million by 2030. To reduce dependence on imports from China, the Indian government has initiated plans and investments in the development and commercialisation of SIBs. Several companies and organisations have invested or are in the process of investing in SIB development. Here are some major players:

Reliance New Energy Solar (RNES). RNES, a part of the Indian conglomerate Reliance, has acquired the UK startup Faradion with a US\$135 million investment. RNES is establishing a Giga-scale manufacturing facility using Faradion's technology, with a production capacity of 1GWh. This acquisition is considered a strategic move to accelerate indigenous SIB production for electric vehicles, mobile applications, and BESS, reducing India's reliance on China. As of June 2023,

RNES is the only company manufacturing SIBs in India.

Sentient Labs. Sentient Labs, a Pune-based research and technology company, is developing sodium-ion batteries that are expected to cost approximately half of what LIBs cost.

Bharat Heavy Electricals Limited (BHEL). BHEL, a Navratna Public Sector Undertaking (PSU) and one of the largest government-owned engineering and manufacturing companies in India, is actively involved in SIB research and development. They have developed prototypes for trials in renewable energy applications.

Amara Raja Batteries. Amara Raja Batteries, based in Hyderabad, is a leading manufacturer of conventional batteries in India. They are exploring the development of sodium-ion batteries for energy storage applications.

Other organisations. Several other organisations are also engaged in SIB technology development in India, including Bharat Energy Storage Technology (BEST), Indo-US MIM Tech Technology, Central Electrochemical Research Institute (CECRI), and Amrita Centre for Nano Sciences and Molecular Medicine (ACNSMM). Additionally, IIT Kharagpur and several startups, such as KPIT Technologies, Indi Energy, and Sodium Energy, are actively working on SIB development. IIT Kharagpur has made advancements in nanomaterials and supercapacitors for next-generation energy storage technologies, with plans for industrial-scale production.

But China dominates the market

China holds a dominant position in the global lithium-ion battery market, accounting for approximately 60% of the market share, with 80% of the



The Giga-scale manufacturing facility being set up by RNES (Credit: <https://www.zigwheels.com>)